DAY 00 / MASTER INDEX

Cloud and LLMOps Learning Index

A guided map of the model, infrastructure, delivery, and operations notes for a production-ready GenAI platform.

14	478	3
TOPIC GUIDES	PDF PAGES	LEARNING PHASES

RECOMMENDED PATH

1 **Understand models and inference**

Days 12-17: ML fundamentals, transformers, LLM behavior, multimodality, model adaptation, and serving.

2 **Build the cloud runtime**

Days 18-20: Docker, Compose, Terraform, AWS infrastructure, Kubernetes, and Helm.

3 **Automate and operate the platform**

Days 21-24: Jenkins CI/CD, Argo CD GitOps, MLflow, environments, governance, and final integration.

HOW TO USE THIS INDEX

Follow the phases in order for a complete learning path, or use the quick locator to jump directly to a model, cloud, deployment, or operations topic.

PHASE 01 / DAYS 12-17

Model and inference foundations

Move from classical machine learning to transformers, LLM behavior, multimodality, model adaptation, and production inference.

DAY 12

ML Fundamentals and Interview Preparation

Builds a practical machine-learning foundation across learning types, evaluation, classical models, NLP, and computer vision.

KEY COVERAGE supervised, unsupervised, and reinforcement learning; train/validation/test splits; leakage; regularization; classification and regression metrics; classical algorithms

PDF 21 pages | **SOURCE** 1,298 Markdown lines

DAY 13

Neural Networks and Transformers

Explains deep-learning mechanics before deriving the transformer architecture and attention mental model.

KEY COVERAGE layers; activation functions; loss; backpropagation; SGD and Adam; self-attention; query, key, and value; multi-head attention; residuals and normalization

PDF 23 pages | **SOURCE** 1,389 Markdown lines

DAY 14

LLM Fundamentals and Model Families

Connects tokenization, training stages, inference behavior, sampling controls, context limits, and major model families.

KEY COVERAGE BPE and SentencePiece; pre-training; instruction tuning; RLHF and DPO; temperature; top-k and top-p; context windows; model-family trade-offs

PDF 21 pages | **SOURCE** 1,195 Markdown lines

DAY 15

Multimodal Models and Diffusion

Introduces systems that combine language with vision and explains how diffusion-based generation works and fails.

KEY COVERAGE vision encoders; adapters and projection layers; cross-attention; multimodal use cases; OCR and spatial limitations; diffusion training and generation

PDF 30 pages | **SOURCE** 1,714 Markdown lines

DAY 16

Training, Fine-Tuning, PEFT and Evaluation

Provides a decision framework for adapting models efficiently, preparing datasets, and evaluating the result.

KEY COVERAGE full fine-tuning; LoRA and QLoRA; continued pre-training; SFT; preference tuning; data cleaning and deduplication; PII removal; evaluation strategy

PDF 28 pages | **SOURCE** 1,591 Markdown lines

DAY 17

LLM Inference, Deployment and Monitoring

Moves models into production through hosted or self-hosted inference, performance optimization, deployment, and operational controls.

KEY COVERAGE API versus self-hosted inference; vLLM, TGI, and Ollama; TTFT and throughput; batching; caching; quantization; streaming; Docker; REST and gRPC; canary rollback

PDF 26 pages | **SOURCE** 1,492 Markdown lines

MEMORY LINE

Understand the model lifecycle before choosing infrastructure: data and training shape capability; inference architecture shapes latency, cost, and reliability.

PHASE 02 / DAYS 18-20

Containers, infrastructure, and runtime

Package the application, declare its AWS foundation, and operate the services through Kubernetes and Helm.

DAY **18A**

Docker and Docker Compose for GenAI

Packages a production-style GenAI application and its dependencies into reliable, observable local services.

KEY COVERAGE images and containers; layers; ports; environment variables; volumes; networks; production Dockerfiles; health checks; graceful shutdown; Compose architecture

PDF 55 pages | **SOURCE** 3,300 Markdown lines

DAY **18B**

Terraform and IaC Fundamentals for AWS

Introduces declarative infrastructure, Terraform language and workflow, state management, and reusable AWS foundations.

KEY COVERAGE providers and resources; variables, locals, and outputs; init, plan, and apply; state; S3 backends; locking; modules; secure configuration

PDF 33 pages | **SOURCE** 1,879 Markdown lines

DAY **19**

AWS GenAI Infrastructure with Terraform

Designs the cloud foundation for a production GenAI SaaS platform using managed AWS services and Terraform.

KEY COVERAGE VPCs and subnets; NAT and endpoints; security groups; EKS; ALB ingress; RDS PostgreSQL; Redis; S3; ECR; IAM; autoscaling

PDF 30 pages | **SOURCE** 1,659 Markdown lines

DAY **20**

Kubernetes and Helm for GenAI Applications

Deploys FastAPI and React services through Kubernetes resources and reusable Helm charts.

KEY COVERAGE Pods, Deployments, Services, and Ingress; ConfigMaps and Secrets; chart structure; values and templates; probes; resources; scaling; rollout safety

PDF 30 pages | **SOURCE** 1,811 Markdown lines

MEMORY LINE Docker packages the workload. Terraform provisions the cloud foundation. Kubernetes runs and scales services. Helm makes deployments reusable.

PHASE 03 / DAYS 21-24

Delivery, GitOps, and LLM operations

Automate testing and release, reconcile deployment state, measure AI behavior, and integrate the final enterprise platform.

DAY 21

Jenkins CI/CD for GenAI

Builds a delivery pipeline that tests, packages, publishes, and deploys a GenAI application to EKS.

KEY COVERAGE controllers and agents; declarative pipelines; lint and tests; RAG golden tests; Docker builds; ECR; Helm deployment; credentials; promotion; rollback

PDF 27 pages | **SOURCE** 1,470 Markdown lines

DAY 22

Argo CD GitOps for GenAI Platforms

Reframes deployment as pull-based reconciliation with Git as the auditable source of truth.

KEY COVERAGE desired state; pull-based delivery; drift detection; CI/CD boundaries; Argo CD components; repository design; synchronization; promotion; self-healing

PDF 55 pages | **SOURCE** 3,141 Markdown lines

DAY 23

MLflow, MLOps and LLMOps

Tracks experiments and production behavior across prompts, traces, evaluations, datasets, feedback, and model promotion.

KEY COVERAGE tracking contracts; parameters, metrics, and artifacts; prompt registry; GenAI tracing; evaluation and scorers; datasets; human feedback; model registry

PDF 46 pages | **SOURCE** 2,546 Markdown lines

DAY 24

Environments and Final End-to-End Integration

Combines the full platform into an enterprise operating model with explicit contracts, ownership, isolation, and lifecycle boundaries.

KEY COVERAGE target architecture; cross-system contracts; monorepo ownership; environment isolation; configuration and secrets; request flow; ingestion; re-indexing; evaluation

PDF 53 pages | **SOURCE** 2,488 Markdown lines

MEMORY LINE

Jenkins proves and packages a change. GitOps reconciles desired state. MLflow measures AI behavior. Environment contracts keep the whole platform governable.

QUICK LOCATOR

Find the right guide fast

Use this cross-reference when you know the capability you need but not the day that covers it.

Need to find...	Start with	Guide
Metrics, leakage, classical ML	Machine-learning fundamentals	**Day 12**
Attention and transformers	Deep-learning architecture	**Day 13**
Tokens, sampling, context, models	LLM behavior	**Day 14**
Vision-language and diffusion	Multimodal generation	**Day 15**
LoRA, QLoRA, SFT, evaluation	Model adaptation	**Day 16**
Serving, latency, caching, rollback	Inference deployment	**Day 17**
Containers and local service stack	Docker and Compose	**Day 18A**
Terraform language and state	Infrastructure as Code	**Day 18B**
VPC, EKS, RDS, Redis, S3, IAM	AWS platform foundation	**Day 19**
Pods, Deployments, Helm charts	Kubernetes runtime	**Day 20**
Build, test, publish, deploy	Jenkins CI/CD	**Day 21**
Drift, reconciliation, promotion	Argo CD GitOps	**Day 22**
Prompts, traces, evaluation, registry	MLflow and LLMOps	**Day 23**
Environments, contracts, ownership	Final platform integration	**Day 24**

PLATFORM MEMORY LINE

Models create capability. Infrastructure provides capacity. Delivery systems move change safely. LLMOps measures behavior. Governance connects every layer.

COLLECTION 14 companion PDFs in cloud/pdfs/ | **TOTAL** 478 PDF pages | **UPDATED** 31 July 2026